IMO Training – Inequalities
Level 2 Assignment 1 Solutions

1. Note that by Cauchy-Schwarz (CS),
$$(a^2+b^2)(1+1) \geq (a+b)^2$$
$$\therefore \sqrt{a^2+b^2} \geq \frac{a+b}{\sqrt{2}}.$$
Similarly, $\sqrt{b^2+c^2} \geq \dfrac{b+c}{\sqrt{2}}$ and $\sqrt{c^2+a^2} \geq \dfrac{c+a}{\sqrt{2}}$.
Adding them up gives the required inequality. Equality holds when $a=b=c$.

2. First observe that as $(a-b)^2+(b-c)^2+(c-a)^2 \geq 0$ for any a, b, c, we have $a^2+b^2+c^2 \geq ab+bc+ca$. Then by CS,
$$\left(\frac{a^3}{b}+\frac{b^3}{c}+\frac{c^3}{a}\right)(ab+bc+ca) \geq (a^2+b^2+c^2)^2$$
$$\left(\frac{a^3}{b}+\frac{b^3}{c}+\frac{c^3}{a}\right) \geq (a^2+b^2+c^2)\frac{a^2+b^2+c^2}{ab+bc+ca},$$
which gives the required inequality. Equality holds when $a=b=c=\dfrac{1}{\sqrt{3}}$.

3. By AM-GM,
$$\begin{aligned} a+\frac{1}{ab-b^2} &= a-b+b+\frac{1}{b(a-b)} \\ &\geq \sqrt[3]{\frac{(a-b)b}{b(a-b)}} \\ &\geq 3. \end{aligned}$$
Equality holds when $a-b=b=\dfrac{1}{b(a-b)}$, i.e. $a=2, b=1$.

4. By CS,

$$((a_1+\frac{1}{a_1})^2+\cdots+(a_n+\frac{1}{a_n})^2)(1+\cdots+1)\geq(a_1+\frac{1}{a_1}+\cdots+a_n+\frac{1}{a_n})^2$$
$$(a_1+\frac{1}{a_1})^2+\cdots+(a_n+\frac{1}{a_n})^2\geq\frac{1}{n}(1+\frac{1}{a_1}+\cdots+\frac{1}{a_n})^2$$

Also, note that by CS,

$$(a_1+\cdots+a_n)(\frac{1}{a_1}+\cdots+\frac{1}{a_n})\geq n^2$$
$$\frac{1}{a_1}+\cdots+\frac{1}{a_n}\geq n^2$$
$$\therefore (a_1+\frac{1}{a_1})^2+\cdots+(a_n+\frac{1}{a_n})^2\geq\frac{(1+n^2)^2}{n}.$$

Equality holds when $a_1=a_2=\cdots=a_n=\frac{1}{n}$.

5. By AM-GM,

$$\begin{aligned}a+b+c&=\frac{a}{3}+\frac{a}{3}+\frac{a}{3}+\frac{b}{2}+\frac{b}{2}+c\\&\geq 6\sqrt[6]{\frac{a^3b^2c}{108}}\\&=\frac{6}{\sqrt[6]{108}}\end{aligned}$$

$\therefore$ The minimum value is $\frac{6}{\sqrt[6]{108}}$.
Equality holds when $\frac{a}{3}=\frac{b}{2}=c$, i.e. $(a,b,c)=(\frac{3}{\sqrt[6]{108}},\frac{2}{\sqrt[6]{108}},\frac{1}{\sqrt[6]{108}})$

6. By AM-GM,

$$\begin{aligned}x^8+y^4+2&=x^8+y^4+1+1\\&\geq 4x^2y\\ \therefore x^8+y^4&\geq 4x^2y-2\end{aligned}$$

Equality holds when $x^8=y^4=1$, i.e. $x=y=1$.

7. Let $x=1-a, y=1-b, z=1-c$. Then $x+y+z=1$, and hence $a=y+z, b=z+x, c=x+y$.
So it is equivalent to show $(x+y)(y+z)(z+x)\geq 8xyz$.
By AM-GM, $x+y\geq 2\sqrt{xy}, y+z\geq 2\sqrt{yz}, z+x\geq 2\sqrt{zx}$.
Multiplying them gives the desired inequality.
Equality holds when $x=y=z=\frac{2}{3}$.

8. By CS,

$$\begin{aligned}
(a^2+b^2)(c^2+d^2) &\geq (ac+bd)^2 \\
(a^2+b^2)^4 &\geq (ac+bd)^2 \\
(a^2+b^2)^2 &\geq ac+bd \\
\therefore \left(\frac{a^3}{c}+\frac{b^3}{d})\right)(ac+bd) &\geq (a^2+b^2)^2 \\
\left(\frac{a^3}{c}+\frac{b^3}{d})\right)(ac+bd) &\geq ac+bd \\
\left(\frac{a^3}{c}+\frac{b^3}{d})\right) &\geq 1
\end{aligned}$$

Equality holds when $\frac{a}{c}=\frac{b}{d}$ and $c^2+d^2=(a^2+b^2)^3$.
Remark: A way to parametrise the equality case is $(a,b,c,d)=(k\cos x, k\sin x, k^3\cos x, k^3\sin x)$, where $k>0, x\in(0,\frac{\pi}{2})$.

9. Claim that $\frac{a^3+b^3+c^3}{a+b+c}\geq\frac{a^2+b^2+c^2}{3}$. Indeed, the inequality is equivalent to

$$\begin{aligned}
3(a^3+b^3+c^3) &\geq (a+b+c)(a^2+b^2+c^2) \\
\iff \frac{3}{2}\sum a^3 &\geq \frac{1}{2}\sum a^3+\sum a^2b \\
\iff \sum a^3 &\geq \sum a^2b
\end{aligned}$$

Which is true by Muirhead Inequality. Similarly, we have

$$\begin{aligned}
\frac{b^3+c^3+d^3}{b+c+d} &\geq \frac{b^2+c^2+d^2}{3} \\
\frac{c^3+d^3+a^3}{c+d+a} &\geq \frac{c^2+d^2+a^2}{3} \\
\frac{d^3+a^3+b^3}{d+a+b} &\geq \frac{d^2+a^2+b^2}{3}
\end{aligned}$$

Adding up gives the desired inequality. Equality holds when $a=b=c=d$.

10. From the condition given, let $a=x+y, b=y+z, c=z+x$, where $x,y,z>0$. (Ravi substitution).
Then the inequality becomes

$$(x+y)^2(y+z)(x-z)+(y+z)^2(z+x)(y-x)+(z+x)^2(x+y)(z-y)\geq 0$$

Expand and simplify gives

$$\begin{aligned}
\sum x^3y+\sum x^2y^2+\frac{3}{2}\sum x^2yz+y^3x+z^3y+x^3z &\geq \sum x^2y^2+\frac{5}{2}\sum x^2yz+y^3z+z^3x+x^3y \\
\iff y^3x+z^3y+x^3z &\geq x^2yz+y^zx+z^2xy
\end{aligned}$$

By CS,

$$\begin{aligned}
(y^3x+z^3y+x^3z)(z^2xy+x^2yz+y^2xz) &\geq (y^2xz+z^2xy+x^2yz)^2 \\
y^3x+z^3y+x^3z &\geq x^2yz+y^zx+z^2xy,
\end{aligned}$$

as desired. Equality holds when $\frac{y}{z}=\frac{z}{x}=\frac{x}{y}$, i.e. $x=y=z$.

11. Homogenise with the condition given and the inequality becomes

$$\begin{aligned}
&7(ab+bc+ca)(a+b+c) \leq 2(a+b+c)^3 + 9abc \\
\iff\ & 7\sum a^2b + \frac{7}{2}\sum abc \leq \sum a^3 + 6\sum a^2b + \frac{7}{2}\sum abc \\
\iff\ & \sum a^2b \leq \sum a^3
\end{aligned}$$

which is true by Muirhead Inequality. Equality holds when $a = b = c = \dfrac{1}{3}$.